1. Content for Tracking_2.docx

Tracking Subsystem Specification

Overview & Purpose

The Tracking Subsystem provides end-to-end visibility into order fulfillment for Buyers and Vendors. It bridges carrier data with internal order states to ensure accurate inventory tracking and automated customer updates throughout the order lifecycle.

Key Workflows

1. Shipment Initialization & Carrier Assignment

- **Vendor Trigger:** When a vendor marks an OrderItem status as **SHIPPED** in their dashboard, the system prompts them to select a carrier (e.g., FedEx, UPS, DHL, USPS) and input a valid tracking number.
- **Data Association:** The system updates the OrderItem record with the tracking_number, carrier_name, and timestamp, transitioning the status from PROCESSING to SHIPPED.
- **Buyer Notification:** An automated email/notification is dispatched to the Buyer containing direct tracking links and estimated delivery dates.

2. Carrier API Synchronization & Webhooks

- **Background Ingestion:** An automated background process periodically polls carrier APIs (or receives inbound carrier webhooks) to capture real-time status changes (In Transit, Out for Delivery, Delivered, or Exception).
- **State Mapping:** Carrier status updates map to internal application statuses:
    - IN_TRANSIT → Item in carrier network.
    - OUT_FOR_DELIVERY → Item in transit to local shipping address.
    - DELIVERED → Item delivery confirmed by carrier.

3. Delivery Confirmation & Review Unlock

- **Status Update:** Upon receiving DELIVERED confirmation from the carrier, the system sets the OrderItem status to DELIVERED.
- **Verified Purchase Status:** Transitioning to DELIVERED enables the **Verified Purchase Review** flag for the Buyer, allowing them to submit product ratings and seller feedback.

Data Model & Attributes

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| ORDER ITEM |

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| item_status : Enum (PENDING, PROCESSING, |

| SHIPPED, DELIVERED, CANCELLED)|

| tracking_number : String (Nullable) |

| carrier_name : String (Nullable) |

| estimated_delivery: DateTime (Nullable) |

| shipped_at : DateTime (Nullable) |

| delivered_at : DateTime (Nullable) |

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2. System Administrator Workflows

System Administrators maintain platform governance, monitor seller compliance, approve merchant accounts, and manage platform-wide fee structures.

Workflow 1: Vendor Store Approval & Onboarding

1. **Application Submission:** When a user registers as a Vendor, their account is placed in a PENDING_APPROVAL state.
2. **Admin Review Interface:** Administrators review business credentials, store names, contact details, and tax identification in the Admin Panel.
3. **Approval Action:**
    - **Approved:** Account status changes to ACTIVE. The Vendor gains access to the Vendor Dashboard to create stores and add products.
    - **Rejected:** Account status changes to REJECTED. An automated email detailing the rejection reasons is dispatched to the applicant.

Workflow 2: Content Moderation & Listing Oversight

1. **Flagging Mechanism:** Buyers or system filters flag listings containing inappropriate content, prohibited items, or intellectual property violations.
2. **Moderation Queue:** Flagged items enter the Admin Moderation Queue with severity tags.
3. **Enforcement Actions:**

- **Delist Product:** Admin toggles is_published = False, hiding the item from public catalog search.
- **Warn Vendor:** Issue an administrative warning via internal messaging.
- **Suspend Store:** Temporary or permanent suspension of a Vendor Store, automatically unpublishing all associated product listings.

Workflow 3: Dispute Resolution & Refund Authorization

1. **Dispute Escalation:** If a Buyer and Vendor cannot resolve an issue (e.g., non-delivery, damaged goods), either party escalates the order item to DISPUTED status.
2. **Evidence Audit:** Administrators review order history, carrier tracking logs, and direct buyer-seller messages.
3. **Resolution Execution:**
    - **Refund Buyer:** Admin triggers a partial or full refund via the Payment Processor API (e.g., Stripe Refund Endpoint).
    - **Release Funds:** Admin dismisses the dispute, authorizing payout release to the Vendor.

Workflow 4: Platform Fee & Commission Management

1. **Fee Configuration:** Administrators configure global or store-specific commission rates (e.g., 5% per transaction) in the Admin Panel.
2. **Payout Calculations:** Upon successful order delivery confirmation, the platform automatically deducts the platform fee from the line-item total prior to vendor payout distribution.